

# Disinformation, Financial Fragility, and Downside Growth Risk

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## Abstract

This paper examines whether a broad disinformation index is associated with downside growth risk and financial fragility. A panel of 175 countries from 2000 to 2025 combines expert-coded Digital Society Project indicators with macroeconomic outcomes. The index measures the political information environment, not the prevalence of false financial messages. Panel local projections estimate a three-year cumulative-growth association of  $-1.52$  percentage points per standard deviation of the index at mean conditioning states. The banking-stress interaction is  $-0.94$ . Country-block bootstrap contrasts do not distinguish the lower-tail slope from the median or upper tail. A chronological forecast comparison provides no evidence of improved downside prediction in the evaluated design. The tenth-quantile forecast-loss gain from adding the index is  $-3.22$  percent; negative gains indicate worse performance. An illustrative financing model gives increasing differences in contamination and fragility. Aggregate index persistence is distinct from individual narrative lifetimes, and a half-life elasticity comparison ranks parameter changes only under explicit sign, cost and effectiveness assumptions. The evidence is observational and does not identify financial-message harm, a crisis threshold or causal returns to information policy.

**JEL codes:** D83, D84, E32, E44, G01, G28, L86.

**Keywords:** disinformation, information pollution, narrative economics, growth-at-risk, financial fragility, local projections.

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# 1 Introduction

Misleading information can affect financial decisions through attention, beliefs and confidence. Evidence on manipulated financial news and social media during bank runs establishes the relevance of information transmission for financial markets (Kogan et al., 2023; Clarke et al., 2021; Cookson et al., 2026). Social-media amplification is not itself evidence that the amplified claims were false. A distinct macroeconomic question is whether an information environment characterized by frequent disinformation is associated with worse growth outcomes, particularly when financial conditions are fragile.

This paper investigates that question with a country panel and an illustrative financing mechanism. Its empirical contribution is a transparent assessment of the information content of a broad disinformation proxy, including conditional mean associations, quantile slopes, fragility interactions and chronological forecast evaluation. It does not estimate the causal macroeconomic effect of financial misinformation or the returns to a particular content intervention.

The mechanism combines signal contamination with an explicitly assumed noise-sensitive lending rule. The resulting investment shortfall has increasing differences in contamination and fragility: deterioration in information can matter more when financing constraints are close to binding. A separate feedback example permits multiple steady states. A scalar persistence calculation gives a conditional comparison of equal proportional changes in model parameters. These results clarify hypotheses and identify assumptions needed for policy analysis; they are not substitutes for measurement or identification.

The empirical proxy combines five DSP indicators of government and party disinformation in 175 countries over 2000–2025. It measures relative intensity in the political information environment, rather than the share of financial messages that are false. The revised analysis retains that distinction throughout. The local projections estimate associations with cumulative outcomes, while separate quantile regressions assess whether slopes differ across the conditional growth distribution. Financial stress is a conditioning state, not an externally assigned treatment.

The revised three-year cumulative-growth coefficient is  $-1.52$  percentage points per standard deviation of the index at mean conditioning states. The corresponding banking-stress interaction is  $-0.94$  points. Two-year conditional growth slopes are  $-0.71$ ,  $-0.55$  and  $-0.82$  points at the tenth, fiftieth and ninetieth percentiles. These numbers must be evaluated with their uncertainty, sample variation and reverse-timing diagnostics, rather than interpreted as treatment effects. Joint bootstrap comparisons do not distinguish the lower-tail slope from the median or upper tail. This weakens the original interpretation of a disproportionately large downside association.

The forecasting extension compares the same macro-financial benchmark with and without the index. It fits component scaling and ranks solely within each training window and respects a one-year predictor-availability lag. Across held-out target years 2015–2024, the change in tenth-quantile pinball loss is reported as a  $-3.22$  percent gain, where negative values mean worse performance. The evaluated design does not support improved downside-risk prediction from adding the index. Latest-vintage source data and a short evaluation period remain material limitations.

**Related literature.** Models of misinformation and narrative formation study platform incentives, social learning and competing explanations (Acemoglu et al., 2024; Mueller-Frank, 2024; Eliaz and Spiegler, 2020). Experimental work documents motivated evaluation and incomplete updating after retractions (Thaler, 2024; Gonçalves et al., 2026). Evidence on political misperceptions and institutional confidence reinforces the need to distinguish a political-information proxy from financial-content exposure (Cariolle et al., 2024; Guriev et al., 2021; Zhuravskaya et al., 2020). Assessments of misinformation also warn against extrapolating from vivid incidents to aggregate harm (Budak et al., 2024; Altay et al., 2023).

The macroeconomic motivation comes from narrative economics (Shiller, 2017, 2019), vulnerable-growth methods (Adrian et al., 2019), and collateral-based amplification (Kiyotaki and Moore, 1997; Bernanke et al., 1999). The feedback illustration is related to uncertainty traps (Fajgelbaum et al., 2017). The empirical tools are local projections (Jordà, 2005) and two-step panel quantile estimation (Canay, 2011). Applying those tools to a persistent observational index does not by itself identify a structural shock. The paper’s contribution lies in the combination of measurement discipline, reproducible conditional evidence, and explicit predictive validation.

Section 2 develops the illustrative mechanism; Section 3 explains measurement; Section 4 presents revised estimation and validation. Sections 5 and 6 delimit risk and policy interpretations, and Section 7 concludes. The appendix provides proofs, robustness results and replication details.

## 2 An Illustrative Mechanism and Its Limits

The model separates an observed proxy for the information environment from financial-message contamination. Its role is to organize hypotheses about conditional associations. The country panel does not identify its structural parameters or a causal response to information policy.

### 2.1 A bounded information-environment proxy

Let  $M_t \in (0, 1)$  denote a normalized information-environment index and  $x_t = \text{logit}(M_t)$ . Consider

$$x_{t+1} = c + \rho x_t + \delta s_t + u_{t+1}. \quad (1)$$

Here  $c$  is an intercept in logit units,  $\rho$  is conditional persistence, and  $s_t$  summarizes other countries’ index values. An intercept can be negative; it is not a count or physical inflow of false messages. The observed index is a rank, so neither  $M_t$  nor  $x_t$  measures the share of financial claims that are false. The process offers a parsimonious description of persistence in an information environment, motivated by narrative economics (Shiller, 2017, 2019).

**Proposition 1** (Scalar dynamics conditional on external inputs). *With  $s_t = 0$ ,  $u_t = 0$  and  $0 < \rho < 1$ , the unique steady state is  $\mu = c/(1 - \rho)$ , and  $x_t = \mu + \rho^t(x_0 - \mu)$ . Consequently  $M_t = \sigma(x_t) \in (0, 1)$  and converges to  $\sigma(\mu)$ . With independent mean-zero innovations of variance  $\sigma_u^2$ , the stationary latent variance is  $\sigma_u^2/(1 - \rho^2)$ . A deviation in the latent state has half-life  $h_{1/2} = \ln(2)/[-\ln(\rho)]$ , exceeding one observation period precisely when  $\rho > 1/2$ .*

This half-life concerns a deviation of the aggregate latent index, not the survival time of a particular narrative. In the stochastic model,  $\mathbb{E}[\sigma(x)]$  generally differs from  $\sigma(\mathbb{E}[x])$ . The nonlinear index also need not inherit the latent state's exact half-life. For jointly evolving countries with a fixed row-stochastic regional matrix  $W$ , the corresponding linear system is  $x_{t+1} = c + (\rho I + \delta W)x_t + u_{t+1}$ . Its stability requires the spectral radius of  $\rho I + \delta W$  to be below one. For nonnegative  $\rho, \delta$ , that radius equals  $\rho + \delta$ . Scalar stability alone therefore does not establish stability of an endogenous regional system.

## 2.2 Contamination and financing constraints

Let  $p \in [0, 1]$  be the unobserved probability that a financial signal is contaminated. A signal about collateral value  $V$  is  $s = V + b\eta d + \varepsilon$ , where  $b \sim \text{Bernoulli}(p)$ ,  $\eta \in \{-1, 1\}$  has equal probabilities,  $d > 0$ , and  $\varepsilon$  has mean zero and variance  $\sigma_\varepsilon^2$ . Assume the disturbances and  $V$  are mutually independent, conditional on  $p$ .

**Lemma 1** (Signal contamination).  $\mathbb{E}[s \mid V, p] = V$  and  $\text{Var}(s \mid V, p) = \sigma_\varepsilon^2 + pd^2$ .

The lemma is about signal noise conditional on the fundamental. It is not a formula for posterior uncertainty  $\text{Var}(V \mid s, p)$ , which also depends on the prior and the signal likelihood. To obtain a transparent financing mechanism, assume directly that lenders apply a haircut increasing in noise variance:

$$L(p) = \lambda\{\bar{V} - \gamma(\sigma_\varepsilon^2 + pd^2)\}, \quad \lambda, \gamma \geq 0. \quad (2)$$

Restrict attention to  $L(p) \geq 0$ ; this is a reduced-form credit rule rather than a derived Bayesian valuation. An entrepreneur with net worth  $W_0$  and desired investment  $I^*$  invests  $\min\{I^*, W_0 + L(p)\}$ . The resulting shortfall can be written

$$H(p, F) = \max\{ap + bF - \kappa_0, 0\}, \quad a, b \geq 0, \quad (3)$$

where  $a = \lambda\gamma d^2$  and  $F$  summarizes financing pressure. This illustration connects noisy information and collateral constraints (Kiyotaki and Moore, 1997; Bernanke et al., 1999; Bond et al., 2012).

**Proposition 2** (Increasing differences). *For  $p_1 \leq p_2$  and  $F_1 \leq F_2$ ,  $H(p_2, F_1) - H(p_1, F_1) \leq H(p_2, F_2) - H(p_1, F_2)$ . If  $ap_2 + bF - \kappa_0 \leq 0$ , the incremental shortfall is zero.*

Linking this mechanism to the data requires an additional hypothesis that financial contamination  $p$  increases with the broad index  $M$ , conditional on other determinants. This mapping is neither observed nor estimated. Increasing differences survive a monotone mapping  $p(M)$ , but the magnitude and a linear interaction specification do not follow from ranks alone. Moreover, Proposition 2 does not establish small mean effects or an ordering of growth-quantile slopes without assumptions about how fragility, constraints and other shocks populate the outcome distribution. Those patterns are empirical questions, not consequences of the proposition by themselves.

### 2.3 Feedback as a theoretical possibility

Suppose a bounded, increasing function  $g : \mathbb{R} \rightarrow \mathbb{R}$  describes stress feedback in latent-state units. Consider

$$\psi(x) = c + \rho x + \varphi g(x), \quad \varphi \geq 0. \quad (4)$$

Writing  $g = G \circ \sigma$  is permissible, but the Lipschitz constant below is that of the complete composite function on the latent scale.

**Proposition 3** (Uniqueness and possible multiplicity). *If  $g$  is globally Lipschitz with constant  $L_g$  and  $0 \leq \rho + \varphi L_g < 1$ ,  $\psi$  has a unique globally attracting fixed point. Multiplicity is possible outside that sufficient condition: with  $c = -3/4$ ,  $\rho = 1/2$ ,  $\varphi = 2$  and  $g(x) = \min\{\max\{x, 0\}, 1\}$ , the fixed points are  $-3/2$ ,  $1/2$  and  $5/2$ , with stable outer points and an unstable middle point.*

Failure of the contraction inequality does not prove a bifurcation. For a smooth parameterized model a fold additionally requires a fixed point with  $\psi'(x) = 1$  and suitable curvature and transversality conditions. The panel does not identify  $\varphi$  or  $g$ , so it cannot locate a country relative to an escalation threshold. The numerical example establishes possibility, not an estimated information trap.

### 2.4 A conditional half-life comparison

**Proposition 4** (Comparative statics of the scalar latent steady state). *For  $c > 0$  and  $0 < \rho < 1$ , the elasticities of  $\mu = c/(1 - \rho) > 0$  are  $\varepsilon_c = 1$  and  $\varepsilon_\rho = \rho/(1 - \rho)$ . Equal infinitesimal proportional reductions in  $\rho$  and  $c$  reduce  $\mu$  by more under the former precisely when  $\rho > 1/2$ . For  $c \geq 0$ ,  $\mu$  is convex in  $\rho < 1$ .*

The common derivative of the logistic transformation cancels when comparing the two marginal reductions in the deterministic index  $\sigma(\mu)$  at the same baseline. This does not imply the same ranking for stochastic welfare or a multi-channel burden score. For  $c = 0$  the proportional elasticity of  $\mu$  is undefined. For  $c < 0$ , reducing  $\rho$  raises  $\mu$  toward zero and raises  $M$  toward  $1/2$ ; a universal pollution-reduction claim is false. The observation period must also be fixed: an annual coefficient cannot be used to rank interventions acting on daily narrative lifetimes.

Let  $\eta_\rho = -d \log \rho / dB_\rho$  and  $\eta_c = -d \log c / dB_c$  denote proportional parameter changes per unit of expenditure, evaluated at positive  $c, \rho$ . Persistence intervention has the larger marginal effect on the latent steady state only if

$$\eta_\rho \frac{\rho}{1 - \rho} > \eta_c. \quad (5)$$

Equal proportional parameter changes are thus an equal-cost comparison only when  $\eta_\rho = \eta_c$ . Neither effectiveness schedule is identified by the country panel. Corrections, friction and authentication can affect more than one parameter, further limiting a one-to-one policy interpretation.

### 3 Data and Measurement

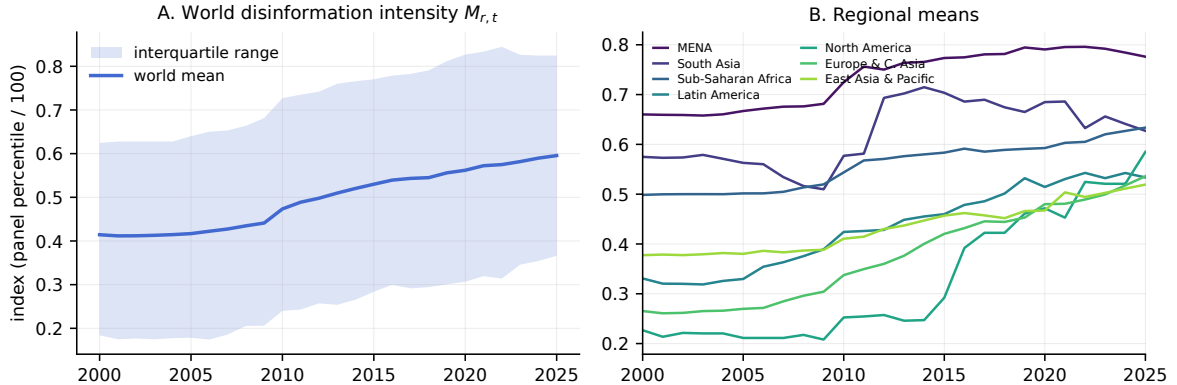
The starting panel is the repository’s frozen merge of the Digital Society Project (DSP) v8, World Development Indicators (WDI), and Worldwide Governance Indicators (WGI). It contains 4,539 country-years for 175 countries in 2000–2025. DSP v8 itself covers 179 countries over those years (Mechkova et al., 2026); this coverage does not imply that every regression contains 175 countries. Observed outcomes and horizon-specific sample sizes are reported below and in the generated tables.

#### 3.1 What the index measures

DSP combines expert assessments through a measurement model (Pemstein et al., 2018). The five components concern false-information dissemination by governments domestically and abroad, political parties domestically and abroad, and foreign governments into the country. Each component is reversed so that higher values indicate more disinformation, then standardized. Weights are respectively  $(1/4, 1/8, 1/4, 1/8, 1/4)$  in that ordering. The weighted composite is transformed into its pooled percentile, using  $(\text{rank} - 1/2)/n$  with average ranks for ties. The resulting  $M_{r,t}$  lies strictly between zero and one.

These are indicators of the broad political information environment. They are not financial-content counts, measured exposure, belief adoption or financial-message accuracy. A value of 0.89 means approximately the 89th percentile of the reference distribution; it does not mean that 89 percent of financial messages are false. No financial-specific validation sample is available in this replication package. Domestic-only or equal weights probe index construction but cannot supply that missing construct validity. The haircut mechanism in Section 2 therefore motivates a hypothesis rather than a measured channel.

The retrospective regressions retain the pooled transformation for comparability. Raw-composite and alternative-weight estimates are reported, rather than assuming that slopes are invariant to monotone transformations. For forecasting, component means, standard deviations and the empirical CDF are re-estimated using each training window only. Evaluation observations outside the training range are clipped to its finite endpoint ranks. The original DSP components required for this procedure are included in the replication data, with source and checksum records.



**Figure 1:** The retrospective disinformation index. The world mean, interquartile range and regional means describe relative positions in a pooled distribution, not estimated shares of false financial messages.

### 3.2 Outcomes and conditioning variables

Real GDP, gross fixed capital formation and export growth are WDI annual percentage changes (World Bank, 2026). Capital formation is labeled *investment*; it does not specifically identify innovation finance. Gini and nonperforming-loan (NPL) ratios provide distributional and banking outcomes. These outcome series are never interpolated or filled in the revised estimation. Growth outcomes sum observed annual rates from  $t$  to  $t + h$ ; this is a cumulative percentage-point sum, not compounded GDP-level growth. Gini and NPL outcomes are observed endpoint changes from  $t - 1$  to  $t + h$ .

Conditioning states are NPLs, trade openness and inequality. Missing NPL conditioning values may use the previous observation for at most two years; Gini conditioning values may use it for at most five years. No future observation is used to fill these states, and a no-fill specification is reported separately. The original repository’s two-sided interpolated columns remain in the frozen input for provenance but are replaced in memory before the revised pipeline runs. Calendar-year matching prevents a gap from being mistaken for a one-year lag.

Controls include lagged GDP growth, log real GDP per capita, CPI inflation clipped to  $[-10, 60]$ , and government effectiveness. Investment and export equations also include the lagged observed outcome; lagged NPL and Gini levels already enter as conditioning states. The frozen WGI vintage ends in 2022 and its existing forward-carried values are retained through 2025; an end-2022 sample check is reported. Historical source releases and contemporaneous publication dates are not reconstructed, so the forecasting exercise uses latest-vintage data and cannot be described as real-time validation.

**Table 1:** Observed data coverage

| Variable             | Mean  | SD    | $n$   |
|----------------------|-------|-------|-------|
| Disinformation index | 0.50  | 0.29  | 4,539 |
| GDP growth           | 3.65  | 5.56  | 4,451 |
| Investment growth    | 5.97  | 42.96 | 3,626 |
| Gini                 | 36.78 | 8.12  | 1,843 |
| Export growth        | 6.01  | 25.42 | 3,800 |
| NPL ratio            | 6.17  | 7.05  | 2,175 |

Unfilled outcome observations. Regression samples are smaller because covariates, lags and leads must be observed.

## 4 Conditional Estimates and Predictive Validation

### 4.1 Estimation and interpretation

The latent-state regression is

$$x_{r,t+1} = \alpha_r + \lambda_t + \rho x_{r,t} + \delta s_{r,t} + \theta_1 F_{r,t} + \theta_2 F_{r,t} x_{r,t} + \theta_3 \pi_{r,t} + u_{r,t+1}, \quad (6)$$

where  $s_{r,t}$  is the leave-one-out regional mean of the latent index. Banking stress is standardized within country in this equation. Regional co-movement is not necessarily transmission: regional economic or political shocks can affect both the index and neighboring countries.

For outcome  $k$  and horizon  $h = 0, \dots, 4$ , the local projection is

$$Y_{r,t,h}^{(k)} = \alpha_r^{(k,h)} + \lambda_t^{(k,h)} + \beta_h^{(k)} M_{r,t}^z + \phi_h^{(k)\top} (M_{r,t}^z W_{r,t}) + \eta_h^{(k)\top} W_{r,t} + \Gamma_h^{(k)\top} Z_{r,t}^{(k)} + e_{r,t,h}^{(k)}. \quad (7)$$

$M^z$  is the standardized index and  $W$  contains standardized NPLs, trade openness and Gini measured at  $t - 1$ .  $Z$  contains clipped inflation, log GDP per capita, government effectiveness, lagged growth and, for investment and exports, the lagged observed outcome. Country and year effects are estimated by iterated within transformation. Standard errors cluster by country. All leads and lags refer to calendar years. For cumulative growth,  $h = 1$  is two years and  $h = 2$  is three years, including the current observation at  $t$ .

The coefficient  $\beta_h$  is the slope at mean conditioning states, not necessarily the population-average marginal effect in the unbalanced estimation sample. The plotted slope at stress  $f$  is  $\beta_h + \phi_{F,h} f$ ; its variance includes  $2f \text{Cov}(\hat{\beta}_h, \hat{\phi}_{F,h})$ . Omitting this covariance is not generally conservative. Country clustering allows within-country dependence but does not eliminate residual regional dependence; a Driscoll–Kraay check is reported in the appendix. Leave-region-out estimates assess sensitivity to regional composition; they are not a substitute for spatially robust inference.

## 4.2 Index persistence

**Table 2:** Conditional index dynamics

| Regressor                    | Estimate | SE      |
|------------------------------|----------|---------|
| Index persistence            | 0.864*** | (0.054) |
| Regional association         | 0.184*** | (0.048) |
| Banking stress               | 0.002    | (0.008) |
| Stress $\times$ latent index | -0.011*  | (0.006) |
| Inflation                    | -0.044   | (0.040) |

Country and year effects; 2,090 observations from 135 countries. Stress is standardized within country. The persistence adjustment discussed in the text is an uncapped approximation.

The revised state equation uses 2090 observations from 135 countries. Estimated persistence is 0.864 (SE 0.054), with a scalar latent-index half-life of 4.7 years conditional on other regressors. Applying the rough Nickell adjustment  $\hat{\rho} + (1 + \hat{\rho})/\bar{T}$  gives 0.985, without capping the result. This approximation is not a validated correction for the full interacted, unbalanced specification. An adjusted value at or above one has no finite stationary half-life under Proposition 1. For comparison,  $\rho = 0.99$  would imply about 69 years, not eight to nine.

The sum of the fitted baseline persistence and regional coefficient is 1.049. In the simplified homogeneous endogenous-network model, a sum above one fails the stability condition. The fitted equation includes fixed effects and stress interactions, so this calculation is a diagnostic, not a complete network estimate. It rules out treating the scalar estimate alone as evidence that the whole regional system is stationary. Neither the index half-life nor its bias adjustment identifies the duration of individual false narratives.



**Table 3:** Conditional associations and banking-stress interactions

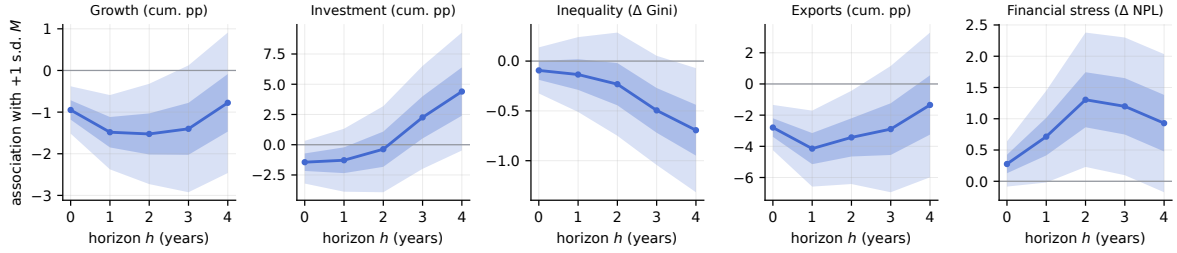
| Outcome    | $h$ | $M$      | SE     | $M \times F$ | SE     | $n$   |
|------------|-----|----------|--------|--------------|--------|-------|
| Growth     | 0   | -0.95*** | (0.35) | -0.14        | (0.17) | 1,735 |
|            | 1   | -1.48*** | (0.54) | -0.45**      | (0.21) | 1,642 |
|            | 2   | -1.52**  | (0.73) | -0.94***     | (0.29) | 1,539 |
|            | 3   | -1.40    | (0.93) | -1.35***     | (0.36) | 1,434 |
|            | 4   | -0.78    | (1.02) | -1.75***     | (0.48) | 1,332 |
| Investment | 0   | -1.44    | (1.07) | 1.06         | (1.30) | 1,628 |
|            | 1   | -1.28    | (1.58) | -0.30        | (1.42) | 1,537 |
|            | 2   | -0.37    | (2.16) | -0.91        | (1.97) | 1,439 |
|            | 3   | 2.26     | (2.59) | -1.44        | (2.40) | 1,338 |
|            | 4   | 4.40     | (2.96) | -2.05        | (2.57) | 1,241 |
| Inequality | 0   | -0.09    | (0.14) | 0.04         | (0.05) | 906   |
|            | 1   | -0.13    | (0.23) | 0.15         | (0.09) | 877   |
|            | 2   | -0.23    | (0.31) | 0.19         | (0.12) | 794   |
|            | 3   | -0.49    | (0.33) | 0.13         | (0.17) | 745   |
|            | 4   | -0.69*   | (0.38) | -0.01        | (0.16) | 673   |
| Exports    | 0   | -2.80*** | (0.88) | -0.25        | (0.45) | 1,661 |
|            | 1   | -4.15*** | (1.48) | -0.70        | (0.59) | 1,567 |
|            | 2   | -3.43*   | (1.82) | -1.64**      | (0.76) | 1,467 |
|            | 3   | -2.90    | (2.46) | -1.69**      | (0.80) | 1,363 |
|            | 4   | -1.35    | (2.82) | -1.69        | (1.10) | 1,263 |
| NPLs       | 0   | 0.28     | (0.22) | 0.23         | (0.20) | 1,663 |
|            | 1   | 0.71     | (0.45) | 0.63**       | (0.29) | 1,562 |
|            | 2   | 1.30**   | (0.65) | 1.06***      | (0.40) | 1,462 |
|            | 3   | 1.20*    | (0.67) | 0.98***      | (0.33) | 1,357 |
|            | 4   | 0.93     | (0.67) | 0.83***      | (0.29) | 1,258 |

Growth, investment and exports sum annual growth from  $t$  through  $t+h$  (percentage points); Gini and NPL outcomes are observed changes from  $t-1$  through  $t+h$ . Thus  $h=2$  includes three annual observations. Country and year effects; country-clustered SEs.  $M$  coefficient is evaluated at mean conditioning states. \*\*\*, \*\*, \* denote 1, 5 and 10 percent significance. No outcome interpolation.

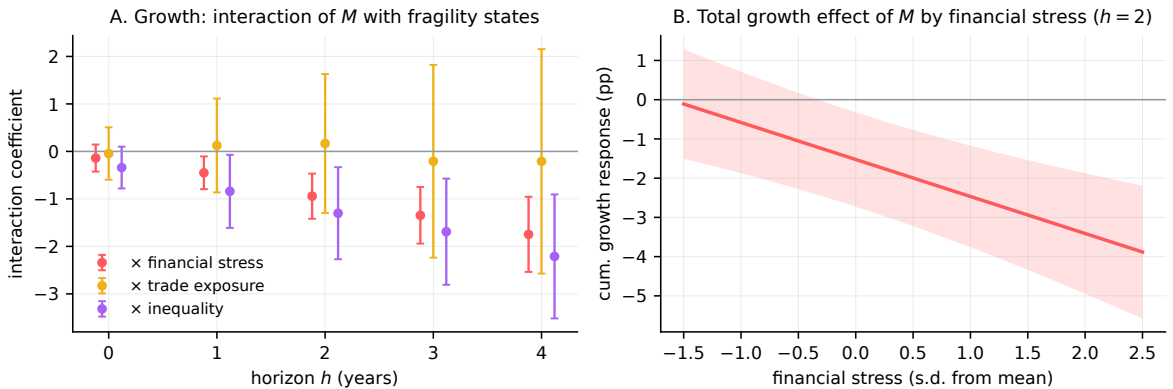
### 4.3 Mean associations and fragility interactions

The three-year growth slope at mean conditioning states is  $-1.52$  percentage points, with a stress interaction of  $-0.94$  points. Table 3 reports uncertainty for every outcome and horizon. These are conditional associations with a persistent index, rather than impulse responses to an exogenous innovation. Contemporaneous macroeconomic controls and lagged conditioning states may themselves respond to disinformation or common shocks, which further prevents a total causal interpretation.

The investment, inequality and NPL estimates are supplementary evidence. A fall in measured inequality is not an increase in inequality harm, an insignificant estimate is not evidence of an exact zero, and average NPL associations cannot establish a stress-feedback mechanism. Interpretation of the financing hypothesis rests on the sign and uncertainty of the growth interaction, its stability across specifications, and the limitations of NPLs as a proxy for financing pressure. At  $h = 2$ , the no-fill growth slope is  $-0.45$  (SE 0.74), while its stress interaction remains negative ( $-1.02$ , SE 0.27). In the post-2015 sample the interaction is  $-0.10$  (SE 0.61), and it is imprecise when Europe and Central Asia are omitted. Thus the baseline evidence does not establish a stable contemporary effect across every sample or conditioning choice.



**Figure 2:** Conditional mean slopes at mean conditioning states. Bands are 50 and 90 percent normal-approximation intervals from country-clustered standard errors. Gini and NPL outcomes use observed endpoints.



**Figure 3:** Growth interactions and the slope conditional on banking stress. The right panel includes the estimated covariance between the index and its stress interaction. Curves describe an estimated regression surface, not a policy treatment response.

#### 4.4 Conditional quantiles and joint uncertainty

For two-year cumulative growth, the first stage fits a two-way fixed-effects mean regression on the index, NPLs, clipped inflation, lagged growth, log GDP per capita and governance. Additive country and year effects are recovered by backfitting and removed from the outcome. The second stage fits pooled quantile regressions on the original covariates and an intercept, following the location-shift logic of [Canay \(2011\)](#). It is not quantile regression of within-demeaned outcomes on within-demeaned regressors.

The assumption that country and year effects are common location shifts across quantiles is substantive and cannot be justified merely because the interest is in slopes. The procedure estimates a conditional distribution, not the outcome that the same country would experience under a treatment. Its covariate set is simpler than the mean-interaction specification; the results are complementary rather than numerically identical estimands.

The revised implementation uses a linear-programming quantile solver and 399 country-block bootstrap draws. Each draw resamples complete country histories with replacement, assigns new identifiers to repeated countries, and re-estimates both the mean-effect and quantile stages. Ninety-percent percentile intervals and paired slope contrasts incorporate the first-stage estimation step. They condition on the constructed index; expert-coding uncertainty and residual cross-country dependence are not fully included. The point estimates

**Table 4:** Two-year cumulative growth: conditional quantile slopes

| Quantile | Slope | 90% interval  | $n$   |
|----------|-------|---------------|-------|
| 10th     | -0.71 | [-1.68, 0.61] | 2,086 |
| 25th     | -0.61 | [-1.58, 0.44] | 2,086 |
| 50th     | -0.55 | [-1.49, 0.44] | 2,086 |
| 75th     | -0.65 | [-1.58, 0.39] | 2,086 |
| 90th     | -0.82 | [-1.81, 0.21] | 2,086 |

399 country-block bootstrap replications, refitting both stages. Intervals condition on the observed index and additive country/year location effects.

**Table 5:** Tests of differences between quantile slopes

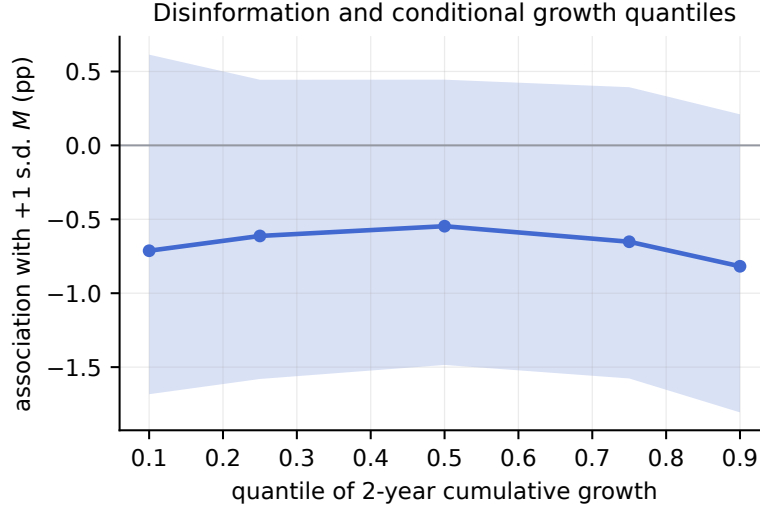
| Contrast        | Difference | 90% interval  | $p$   |
|-----------------|------------|---------------|-------|
| 10th minus 50th | -0.17      | [-0.58, 0.48] | 0.575 |
| 10th minus 90th | 0.11       | [-0.44, 0.92] | 0.805 |

Paired country-block draws preserve covariance between quantile estimates;  $p$  values use the centered bootstrap null distribution. These are pointwise tests, without multiplicity adjustment.

at the tenth, median and ninetieth quantiles are  $-0.71$ ,  $-0.55$  and  $-0.82$ . Whether the lower-tail slope is distinguishable from the center or upper tail is assessed directly in [Table 5](#). The paired tests do not reject equal slopes (tenth versus median,  $p = 0.575$ ; tenth versus ninetieth,  $p = 0.805$ ). All reported quantile slope intervals include zero. The earlier claim of a statistically distinct, roughly threefold lower-tail association is therefore withdrawn.

#### 4.5 Reverse timing and identification limits

Country and year effects remove additive persistent differences and common year shocks. They do not remove time-varying institutional deterioration, conflict, political instability or reverse



**Figure 4:** Conditional quantile slopes with 90 percent country-block bootstrap intervals. Both estimation stages are refitted in each draw.

causality. Associations with prior outcomes are reported in Appendix B. Because the index is persistent and lagged-outcome controls overlap the placebo windows, these reverse-timing regressions are diagnostics, not experimental pre-trend tests. In particular, controlling for a dependent variable’s own overlapping lag cannot establish the absence of confounding.

The previous draft’s regional-pressure instrument interacted with internet penetration does not establish identification. Internet exposure may affect growth through other channels; regional pressure may reflect regional shocks; and 2000–2004 exposure is not pre-period for a sample beginning in 2000. Weak-IV estimates with large uncertainty cannot validate the sign. The revised paper withdraws that evidence rather than treating it as causal corroboration. The repository retains the original results in a clearly marked legacy directory for audit, with the optional IV implementation restricted to post-2004 observations and its excluded-instrument diagnostic corrected. It is not part of the revised empirical evidence.

#### 4.6 Chronological forecast evaluation

At each annual origin  $t = 2014, \dots, 2023$ , the exercise predicts GDP growth in  $t + 1$  using covariates from  $t - 1$ . Training targets end no later than  $t - 1$ . This conservative timing prevents use of current-year realized growth or a future outcome in model fitting. Each model uses the same observed sample and country intercepts; no future year effect is estimated.

The benchmark contains growth, clipped inflation, log GDP per capita, trade openness, NPLs, private credit and government effectiveness. Missing NPL and credit predictors can be carried forward for at most two years, never backward. The augmented model adds the DSP index. All five component standardizations, the rank transform, and regression scaling are fitted anew using training observations only. Models estimate the conditional mean and the tenth, median and ninetieth quantiles directly. The benchmark and augmented models use identical training and evaluation samples at each origin.

The tenth-quantile loss gain is  $-3.22$  percent and the mean-squared error gain is  $-0.14$  percent. Negative gains denote deterioration. None of the four loss metrics improves at the

**Table 6:** Pseudo out-of-sample forecast comparison

| Loss               | Benchmark | With index | Gain (%) | 90% interval   |
|--------------------|-----------|------------|----------|----------------|
| Pinball 0.1        | 1.289     | 1.331      | -3.22    | [-5.44, 1.29]  |
| Pinball 0.5        | 1.535     | 1.541      | -0.41    | [-1.57, 0.41]  |
| Pinball 0.9        | 0.884     | 0.904      | -2.30    | [-5.04, -0.51] |
| Mean squared error | 23.440    | 23.474     | -0.14    | [-0.53, 0.43]  |

Positive gain means lower loss. Identical evaluation samples; target years 2015–2024. Predictor availability lag is one year; training targets end before the forecast origin. Components, index ranks and scaling are fitted within training windows. Intervals resample entire target years (1,999 draws; only ten year blocks). Latest-vintage inputs make this a pseudo rather than real-time evaluation.

point estimate; the tenth-quantile gain interval includes zero. These results do not establish operational forecasting value for this index and benchmark. Country-specific intercepts with sparse training histories and omitted common-shock predictors may also limit performance. Intervals resample complete target years, preserving dependence among countries in a given year, but ten target years provide limited precision and do not fully address serial dependence across years. The exercise is pseudo out of sample: source vintages may contain revisions, expert scores may be retrospective, and their actual publication calendars are not reconstructed. A useful association in the panel therefore need not imply an operational early-warning improvement.

## 5 What Can Be Quantified as Risk?

The defensible quantitative outputs are conditional growth profiles, uncertainty intervals, observed banking-stress interactions, and held-out forecast losses. At an observed stress value  $f$ , the reported profile is  $\hat{\beta}_h + \hat{\phi}_{F,h}f$ , with other conditioning states at their reference values. It should be evaluated within the empirical support and with its covariance-based uncertainty. A change in this profile is not an estimated intervention effect.

The earlier twelve-year growth scenarios are withdrawn. The original code convolved cumulative local-projection coefficients with repeated deviations in the index level and divided the result by  $\sqrt{H}$ . Neither overlapping cumulative coefficients nor this scaling identifies annual growth responses to a sequence of structural shocks. Differencing cumulative coefficients alone would not repair the additional problem that the regressor is a persistent, endogenous level rather than an identified innovation. A replacement scenario system would require explicit dynamic equations, shock identification, and validation of the mapping from interventions to parameters. Those ingredients are not supplied by the current panel.

The composite burden score is also withdrawn from quantitative claims. Normalizing a channel by its estimated two-year response is unstable when that response is near zero and uncertain. Reversals of the previous stability-only ranking cannot be dismissed as an uninformative case while claiming a general ordering. A bounded mathematical score does not establish welfare meaning, comparability, or reliable policy rankings. The appendix retains the elementary boundedness result solely to document the mathematical scope of the original construction. No GDP-loss totals, burden reductions, or empirical trap thresholds are inferred from it.

## 6 Policy Interpretation and Research Requirements

The results motivate scrutiny of the information environment alongside financial fragility, subject to the predictive validation in Table 6. They do not establish that collecting the proposed index improves decisions, that a financial rumor persists for the aggregate index’s half-life, or that a measured country is near a crisis threshold.

Rapid official communication, message authentication and restrictions on content propagation are distinct candidate interventions. Their effects may span the model intercept, persistence, innovation variance, public trust and the reach of accurate information. Ranking them requires estimates of these responses, implementation costs and possible adverse effects. The conditional elasticity result in Proposition 4 does not provide those estimates. Equation (5) makes the missing comparison explicit.

A stronger policy study would link independently identified financial falsehoods to audience exposure, bank funding or lending outcomes and credible correction timings. It would then estimate the response to an intervention using a design with a defensible counterfactual, rather than relabeling a country-level association. A financial-content validation sample would also test whether the broad DSP proxy captures the mechanism proposed here. Out-of-time and real-time validation against richer financial-condition, uncertainty and geopolitical-risk benchmarks would be needed before proposing an operational monitoring instrument. These are requirements for stronger claims, not results claimed by the present paper.

## 7 Conclusion

A broad disinformation index can be examined as a correlate of macroeconomic risk without equating political-information scores with financial-message contamination. The revised panel estimates a three-year cumulative-growth slope of  $-1.52$  percentage points at mean conditioning states and a banking-stress interaction of  $-0.94$ . Quantile inference and chronological prediction evaluate the scope of the evidence, including the possibility that in-sample associations do not improve held-out forecasts.

The illustrative financing model supplies an increasing-differences result under an explicitly assumed lending rule. Feedback can generate multiple steady states for some parameters, but the empirical exercise does not identify a trap threshold. Persistence comparative statics rank parameter changes only under sign and effectiveness restrictions; they do not identify policy cost-effectiveness. The remaining contribution is a reproducible conditional analysis, with measurement, identification and forecasting limitations made explicit. Financial-specific data and credible intervention variation are the next steps needed to support causal policy conclusions.

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## A Proofs and Formalization Scope

### A.1 Scalar state dynamics

The fixed-point equation  $x = c + \rho x$  gives  $\mu = c/(1 - \rho)$ . Subtracting it from the recursion yields  $x_{t+1} - \mu = \rho(x_t - \mu)$ , which proves the closed form and convergence for  $|\rho| < 1$ . Logistic bounds and strict monotonicity follow from  $e^{-x} > 0$ . With independent innovations, the stationary representation is  $x_t = \mu + \sum_{j \geq 0} \rho^j u_{t-j}$  and the variance follows by a geometric sum. Solving  $\rho^h = 1/2$  gives the latent half-life formula. These statements condition on absent or fixed external inputs; they are not a stationarity proof for the fitted regional panel.

For fixed nonnegative row-stochastic  $W$ , the vector of ones is an eigenvector with eigenvalue one. Hence  $\rho + \delta$  is an eigenvalue of  $A = \rho I + \delta W$ . The induced infinity norm is  $\rho + \delta$ , so the spectral radius is exactly that value for nonnegative  $\rho, \delta$ . A time-varying network or country-dependent interaction coefficients require a separate stability analysis.

### A.2 Signal noise and the lending assumption

Independence gives  $\mathbb{E}[b\eta] = 0$  and  $\text{Var}(b\eta d) = d^2 \mathbb{E}[b\eta^2] = pd^2$ . Adding independent  $\varepsilon$  proves Lemma 1. The posterior distribution of  $V$  has not been specified; the lemma does not compute its variance. Under the assumed credit rule (2),

$$I^* - \min\{I^*, W_0 + L(p)\} = \max\{\lambda\gamma d^2 p + I^* - W_0 - \lambda\bar{V} + \lambda\gamma\sigma_\varepsilon^2, 0\}.$$

Collecting the financing-pressure terms gives (3). This is a conditional derivation from a stated reduced-form lending rule, not a Bayesian microfoundation.

### A.3 Increasing differences

Let  $g(z) = \max\{z, 0\}$ ,  $s = ap_1 + bF_1 - \kappa_0$ ,  $u = a(p_2 - p_1) \geq 0$  and  $v = b(F_2 - F_1) \geq 0$ . The increment  $g(s + u) - g(s)$  is nondecreasing in  $s$ , which implies  $g(s + u) - g(s) \leq g(s + v + u) - g(s + v)$ . This proves increasing differences. When  $ap_2 + bF - \kappa_0 \leq 0$ , both terms of the contamination increment are zero. No quantitative claim about mean or quantile growth effects follows without further assumptions on aggregation and other shocks.

### A.4 Feedback and multiplicity

The map in (4) is globally Lipschitz with constant at most  $\rho + \varphi L_g$ . When this bound is below one, Banach's fixed-point theorem on the complete space  $\mathbb{R}$  gives existence, uniqueness and global convergence. In the piecewise-linear example, solving on each of  $x \leq 0$ ,  $0 \leq x \leq 1$  and  $x \geq 1$  gives respectively  $-3/2$ ,  $1/2$  and  $5/2$ . Slopes at those roots are  $1/2$ ,  $5/2$  and  $1/2$ . The example proves that multiplicity is possible, not that it begins when an upper bound on a derivative reaches one.

## A.5 Elasticities and the optional burden bound

For  $c > 0$  and  $0 < \rho < 1$ ,

$$\frac{\partial \mu}{\partial c} = \frac{1}{1 - \rho}, \quad \frac{\partial \mu}{\partial \rho} = \frac{c}{(1 - \rho)^2}, \quad \frac{\partial^2 \mu}{\partial \rho^2} = \frac{2c}{(1 - \rho)^3}.$$

Multiplying the first derivatives by  $c/\mu$  and  $\rho/\mu$  establishes Proposition 4. The same algebraic elasticity identity extends to  $c \neq 0$ , but negative  $c$  reverses the desired direction of a reduction in persistence. At  $c = 0$ , division by  $\mu$  is undefined. Applying the chain rule to expenditure gives (5).

For completeness, if  $M_h \in [0, 1]$ ,  $|Y_h| \leq \bar{Y}$  and  $\kappa > 0$ , then  $D_H = \sum_{h=0}^H (1 + \kappa)^{-h} M_h Y_h$  satisfies  $|D_H| \leq \bar{Y}(1 + \kappa)/\kappa$  by the triangle inequality and the geometric-series bound. Signed outcomes permit negative scores. This bound is not a welfare result and does not validate the withdrawn normalization.

## A.6 What the Lean files establish

The repository includes proofs of logistic bounds and monotonicity; scalar deterministic dynamics and a geometric variance identity; increasing differences of the max-linear harm function; uniqueness given a contraction bound and an explicit three-root example; an absolute burden bound; and algebraic derivative, elasticity and midpoint-convexity results. These are selected mathematical statements, not a machine verification of all economic interpretations. In particular, the files do not establish construct validity, posterior-risk microfoundations, empirical identification, forecast performance, intervention effectiveness, stochastic ergodicity of the full panel, or a data-calibrated bifurcation threshold. The formalization mapping and build instructions are recorded in the repository.

## B Additional Empirical Evidence

**Table 7:** Associations with prior outcomes

| Outcome    | $h$ | Coefficient | SE     | $n$   |
|------------|-----|-------------|--------|-------|
| Growth     | -2  | -1.18**     | (0.54) | 1,734 |
| Growth     | -1  | -0.73**     | (0.33) | 1,735 |
| Investment | -2  | -2.08       | (1.96) | 1,630 |
| Investment | -1  | -2.13*      | (1.25) | 1,638 |
| Inequality | -2  | 0.14        | (0.21) | 987   |
| Inequality | -1  | 0.12        | (0.13) | 958   |
| Exports    | -2  | -2.63**     | (1.12) | 1,662 |
| Exports    | -1  | -2.76***    | (0.92) | 1,671 |
| NPLs       | -2  | 0.24        | (0.59) | 1,488 |
| NPLs       | -1  | 0.17        | (0.28) | 1,594 |

Past growth outcomes sum years  $t + h$  through  $t - 1$ ; past level changes run from  $t + h - 1$  to  $t - 1$ . Lagged outcomes are omitted to avoid mechanical overlap. These are reverse-timing diagnostics, not clean falsification tests for a persistent index.

Alternative-weight specifications reconstruct the index from the published components. The raw-composite check removes the empirical-CDF transformation. Sample restrictions,

**Table 8:** Growth associations under alternative specifications ( $h = 2$ )

| Specification                   | $M$      | SE     | $M \times F$ | $n$   |
|---------------------------------|----------|--------|--------------|-------|
| dk                              | -1.52*** | (0.39) | -0.94**      | 1,539 |
| domestic_only                   | -1.56**  | (0.70) | -0.88***     | 1,539 |
| end2022                         | -0.78    | (0.84) | -0.72**      | 1,232 |
| equal_weights                   | -1.43*   | (0.75) | -0.99***     | 1,539 |
| index_history                   | -2.44*** | (0.77) | -0.96***     | 1,539 |
| no_fill                         | -0.45    | (0.74) | -1.02***     | 1,041 |
| Omit East Asia & Pacific        | -1.57**  | (0.78) | -1.05***     | 1,361 |
| Omit Europe & Central Asia      | -1.69    | (1.20) | -0.90        | 879   |
| Omit Latin America & Caribbean  | -2.28*** | (0.79) | -0.93***     | 1,297 |
| Omit Middle East & North Africa | -1.68**  | (0.72) | -1.00***     | 1,447 |
| Omit North America              | -1.63**  | (0.75) | -0.92***     | 1,507 |
| Omit South Asia                 | -1.27*   | (0.72) | -0.91***     | 1,484 |
| Omit Sub-Saharan Africa         | -0.67    | (0.77) | -0.85***     | 1,259 |
| post2015                        | -1.51*   | (0.81) | -0.10        | 839   |
| raw_composite                   | -1.45*   | (0.83) | -0.94***     | 1,539 |

No-fill uses observed conditioning states only. Raw-composite uses the unranked weighted index; each index is standardized in its specification. Alternative weights and samples can change results; stability is not presumed.

no-fill conditioning, a lagged-index control, Driscoll–Kraay inference and leave-region-out estimates are explicitly recomputed. Their signs, uncertainty and sample sizes should be compared with the baseline rather than summarized as uniformly robust. The Driscoll–Kraay calculation applies a Bartlett kernel with bandwidth two to year-aggregated within-regression scores; inference uses year-count minus one degrees of freedom. The short time dimension limits this approximation. A lagged-index specification asks whether the current index adds information conditional on its recent history, which is a different estimand from the baseline level association.

The Monte Carlo exercise simulates the repository’s own stylized linear-interaction data-generating process in 200 replications. Its output is retained as a numerical estimator diagnostic. It does not test the DSP measurement process, time-varying confounding, the revised forecasting exercise or the assumed financing mechanism. Bias correction reduces but need not eliminate dynamic-panel bias.

## C Replication and Data Provenance

The public repository is <https://github.com/jameskrice7/Financial-Disinformation-as-a-Macroeconomic-Risk>. The revised source, figures and tables are versioned with the empirical code. The entry point `code/scripts/run_all.py` uses the frozen merged panel and a separately documented extraction of DSP v8 components. It regenerates state and local-projection estimates, all reported robustness checks, quantile bootstrap draws and contrasts, held-out predictions and losses, Monte Carlo diagnostics, figures, table environments and numerical macros used in the text. Source hashes and software versions accompany the outputs. The main analysis runs without downloading a new WDI vintage.

The frozen panel contains observed and historically interpolated columns. The revised pipeline preserves observed outcomes, replaces conditioning columns with limited forward

carries, and fits forecasting transformations within training windows. Data coverage is not identical to complete-case regression coverage. A separate provenance record identifies source URLs, retrieval information available for this revision, and the limitations of the older merged panel. Its original raw-vintage details cannot be reconstructed from an undocumented download date, so the paper makes no real-time claim.

Legacy scenario, IV and burden outputs are retained only for audit under `output/legacy`. They do not feed the revised paper. Numerical tests cover calendar alignment, observed outcomes, training-only transformations, bootstrap contrasts and covariance propagation. Source changes and remaining research limitations are documented in the revision note.